

Customer Lifetime Value Predictive Framework

BTYD Probabilistic Models · Machine Learning Regression · Marketing Strategy

STUDENT

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ACADEMIC YEAR

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COURSE

B.Sc. in Artificial Intelligence

DATASET

Brazilian E-Commerce Public Dataset by Olist

Financial Planning & Budget Forecasting

75,319

Calibration Customers

customers with purchase history

R\$ 66,572

Actual Holdout Revenue

realized in the 90-day holdout period

R\$ 88,966

BTYD Predicted CLTV

90-day probabilistic forecast

R\$ 13,758

ML Predicted Revenue

Random Forest · test set (20%)

MODEL COMPARISON — 90-Day Holdout (full calibration base)

Model	MAE	MSE	RMSE	Total Predicted	Gap vs Actual
BTYD (BG/NBD + Gamma-Gamma)	1.90	231.56	15.22	R\$ 17,599*	+ R\$ 6,195
Random Forest (ML Regression) ✓	1.65	218.21	14.77	R\$ 13,758*	+ R\$ 2,354

* test set (20% of calibration customers, ~15,064 customers)

RECOMMENDATION FOR BUDGET FORECASTING

Use Random Forest for short-term budget forecasting: it achieves lower MAE (1.65 vs 1.90), MSE (218.21 vs 231.56) and RMSE (14.77 vs 15.22), and estimates total quarterly revenue **R\$ 2,354 closer to actual** than BTYD. BTYD overestimates by R\$ 6,195 because it assigns non-zero purchase probability to every "alive" customer, most of whom do not transact in the quarter.

Operational Marketing & Customer Targeting

93,357

Total customers
observed (full dataset)

3.0%

Repeat-customer rate
(lifetime)

0.56%

Calibration customers
active in holdout

421

Customers who actually
returned in 90 days

BTYD-BASED CUSTOMER SEGMENTATION (calibration period signals)

WIN-BACK

High CLTV · Low p(alive)

Customers who purchased multiple times during calibration but did not return in holdout. High past value, signs of churn.

ACTION

Personalized reactivation email, time-limited discount coupon, cross-sell based on past categories.

RETAIN

High CLTV · High p(alive)

Top-value customers still considered active by BG/NBD. Highest expected future contribution.

ACTION

Loyalty program enrolment, early access to deals, VIP customer service, subscription nudge.

NURTURE

Low CLTV · High p(alive)

Recent first-time or low-frequency buyers with signs of life. Growth potential if engaged early.

ACTION

Second-purchase incentive (free shipping), category recommendations, triggered post-purchase flow.

MONITOR

Low CLTV · Low p(alive)

One-time buyers inactive for a long period. High cost-to-recover vs low expected return.

ACTION

Exclude from paid campaigns. Include only in low-cost re-engagement flows (push notification, newsletter).